Printed Pages: 01 **Sub Code:BOP-482**

Paper Id: 237059 **Roll No.** | | | | | | | | | | |

B. PHARM
(SEM VIII) THEORY EXAMINATION 2022-23
PHARMACEUTICS-X (PHARMACEUTICAL BIOTECHNOLOGY)

Time: 3 Hours ***Total Marks: 70***

Note: Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt *all* questions in brief. 2 x 7 = 14

(a) What do you mean by monoclonal antibodies?
(b) Define antigen and haptens.
(c) Write the name of two microorganisms from which penicillinase enzyme obtained.
(d) Write the name of cells involved in cellular immunity.
(e) Write the name of enzymes mainly involved in r-DNA technology.
(f) Define bacteriostatic and bactericidal antibiotics.
(g) Describe the term vaccines and vaccination.

SECTION B

2. Attempt any *three* of the following: 7 x 3 = 21

(a) Discuss different types of immunity and cells involved in body's immune system.
(b) Explain the role of rDNA technology in vaccine production.
(c) Discuss antigen-antibody reaction and their pharmaceutical application.
(d) Explain methods used for mutants isolation.
(e) Write a detail note on production of enzyme amylase from fermentation method.

SECTION C

3. Attempt any *one* part of the following: 7 x 1 = 7

(a) Explain the principle of selection and design of fermenters.
(b) Discuss the role of the aeration system and its components in fermenter working.

4. Attempt any *one* part of the following: 7 x 1 = 7

(a) Explain in detail different types of enzyme immobilization methods with examples.
(b) Discuss applications of enzyme immobilization in pharmaceutical industry.

5. Attempt any *one* part of the following: 7 x 1 = 7

(a) Discuss hybridoma technology and steps involved in it with diagram.
(b) Explain the methods used for antibiotics producing microorganisms from soil.

6. Attempt any *one* part of the following: 7 x 1 = 7

(a) Write the applications of monoclonal antibodies in pharmaceuticals.
(b) Explain microbial biotransformation reactions in reference to steroids production.

7. Attempt any *one* part of the following: 7 x 1 = 7

(a) Discuss production method and application of insulin.
(b) Discuss production method and application of Somatotropin.

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